Script is made to optimize power delivery and thermals on my T14s GEN 1 using RYZEN 5 PRO 4650u when on BAT & AC.

Please, do yourself a favor and read script before execution.

DO NOT RUN SCRIPTS BLINDLY!

2. Post-Installation: Granting SELinux Permissions

Because Fedora uses strict SELinux security policies by default, it actively blocks background services from accessing raw hardware memory. To allow the power profiles to apply automatically in the background, you must grant the systemd services "unconfined" permissions.

Copy and paste the following block into your terminal to configure the permissions and start the service:

```
# 1. Grant unconfined SELinux access to the AC profile service sudo sed -i '/\
[Service\]/a SELinuxContext=system_u:system_r:unconfined_t:s0'
/etc/systemd/system/ryzenadj-ac.service

# 2. Grant unconfined SELinux access to the Battery profile service sudo sed -i
'/\[Service\]/a SELinuxContext=system_u:system_r:unconfined_t:s0'
/etc/systemd/system/ryzenadj-battery.service

# 3. Reload systemd to apply the new configurations sudo systemctl daemon-reload

# 4. Trigger the AC service to verify the permissions sudo systemctl start
ryzenadj-ac.service
```

Example:

```
mk@fedora-m-001:~/Downloads$ # 1. Add the unconfined SELinux context to the AC service
sudo sed -i '/\[Service\]/a SELinuxContext=system_u:system_r:unconfined_t:s0' /etc/systemd/system/ryzenadj-ac.service

# 2. Add the unconfined SELinux context to the Battery service
sudo sed -i '/\[Service\]/a SELinuxContext=system_u:system_r:unconfined_t:s0' /etc/systemd/system/ryzenadj-battery.service

# 3. Reload systemd to recognize the changes
sudo systemctl daemon-reload

# 4. Trigger the AC service again to verify it works
sudo systemctl start ryzenadj-ac.service
mk@fedora-m-001:~/Downloads$ sudo journalctl -u ryzenadj-ac.service -n 15 --no-pager
avg 03 22:08:31 fedora-m-001 ryzen-profile-ac[50329]: SMU_SERVICE REQ_ID:0x15
avg 03 22:08:31 fedora-m-001 ryzen-profile-ac[50329]: SMU_SERVICE REQ: arg0: 0x61a8, arg1:0x0, arg2:0x0, arg3:0x0, arg4: 0x0, arg5: 0x0
avg 03 22:08:31 fedora-m-001 ryzen-profile-ac[50329]: SMU_SERVICE REP: REP: 0x1, arg0: 0x61a8, arg1:0x0, arg2:0x0, arg3:0x0, arg4: 0x0, arg5: 0x0
avg 03 22:08:31 fedora-m-001 ryzen-profile-ac[50329]: SMU_SERVICE REQ_ID:0x16
avg 03 22:08:31 fedora-m-001 ryzen-profile-ac[50329]: SMU_SERVICE REQ: arg0: 0x55f0, arg1:0x0, arg2:0x0, arg3:0x0, arg4: 0x0, arg5: 0x0
avg 03 22:08:31 fedora-m-001 ryzen-profile-ac[50329]: SMU_SERVICE REP: REP: 0x1, arg0: 0x55f0, arg1:0x0, arg2:0x0, arg3:0x0, arg4: 0x0, arg5: 0x0
avg 03 22:08:31 fedora-m-001 ryzen-profile-ac[50329]: SMU_SERVICE REQ_ID:0x19
avg 03 22:08:31 fedora-m-001 ryzen-profile-ac[50329]: SMU_SERVICE REQ: arg0: 0x5a, arg1:0x0, arg2:0x0, arg3:0x0, arg4: 0x0, arg5: 0x0
avg 03 22:08:31 fedora-m-001 ryzen-profile-ac[50329]: SMU_SERVICE REP: REP: 0x1, arg0: 0x5a, arg1:0x0, arg2:0x0, arg3:0x0, arg4: 0x0, arg5: 0x0
avg 03 22:08:31 fedora-m-001 ryzen-profile-ac[50329]: Successfully set stapm_limit to 22000
avg 03 22:08:31 fedora-m-001 ryzen-profile-ac[50329]: Successfully set fast_limit to 25000
avg 03 22:08:31 fedora-m-001 ryzen-profile-ac[50329]: Successfully set slow_limit to 22000
avg 03 22:08:31 fedora-m-001 ryzen-profile-ac[50329]: Successfully set tctl_temp to 90
avg 03 22:08:31 fedora-m-001 systemd[1]: ryzenadj-ac.service: Deactivated successfully.
avg 03 22:08:31 fedora-m-001 systemd[1]: Finished ryzenadj-ac.service - Apply RyzenAdj AC Power Profile.
mk@fedora-m-001:~/Downloads$ ^C
```

Verifying the Setup

To confirm that SELinux is no longer blocking the script, check the systemd logs:

```
sudo journalctl -u ryzenadj-ac.service -n 15 --no-pager
```

What to look for: If the configuration was successful, the log will show that the limits were applied and end with a Deactivated successfully or code=exited, status=0/SUCCESS message instead of throwing an exception.

Check screenshot in the example above.

HOW TO's

How to make script executable?

```
chmod +x SCRIPTNAME.sh
```

How to run?

```
./SCRIPTNAME.sh
```